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Pioneering Green Steel in MENA: Hydrogen Pilot Project for Industrial Decarbonization

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Abstract

Steel manufacturing accounts for approximately 7% of global energy-related CO₂ emissions, making it a priority sector for deep decarbonization (IEA, 2023). This paper describes a pilot project that produces green hydrogen via proton exchange membrane (PEM) electrolysis integrated into a commercial-scale direct iron reduction (DRI) plant owned by a leading steel manufacturing company based in Abu Dhabi, UAE. The pilot project represents the first industrial hydrogen-based direct reduced iron (H₂-DRI) capacity in the Middle East and North Africa (MENA), in which natural gas is replaced by green hydrogen as the reducing agent. Green hydrogen offers a carbon-free reducing agent, generating water vapor rather than CO₂ in the iron ore reduction process (Nurdiawati & Urban, 2021), and has been identified as a key pathway in international initiatives such as the Hydrogen Breakthrough Agenda (UNEPFI, 2021). The pilot project aligns with the UAE Hydrogen Strategy 2050, which targets 1.4 million tonnes of annual hydrogen production by 2031 (U.ae, 2025a) and supports the national Net Zero by 2050 commitment (U.ae, 2025b). Operationally, a pressurized PEM electrolyser supplied up to 42 kg H₂/h at >99.99 vol % H₂ purity to the DRI reactor, achieving metallization rates of 97.2% while reducing process CO₂ emissions by ~90% to the natural-gas- based DRI baseline when the direct-reduction plant operates on 100% hydrogen as the reducing gas. Lifecycle assessment following ISO 14040/44 guidelines indicated a 99% reduction in cradle-to-gate emissions when powered entirely by solar energy. The pilot project gained practical experience in producing green hydrogen through electrolysis with PEM technology and in the process understanding the engineering and operational challenges involved during execution. Technical challenges—including hydrogen embrittlement, manifold flow distribution, and refractory wear—were mitigated through advanced coatings, computational fluid dynamics (CFD)-optimized manifolds, and high-zirconia materials. The project demonstrated technical readiness, safety integrity, and compatibility with existing DRI infrastructure, while creating early market demand for green steel through downstream agreements with the UAE construction sector. These findings confirm that hydrogen-based steelmaking is not only viable under MENA climatic and industrial conditions but also strategically positions the UAE as a leader in low-carbon materials, which have the potential to satisfy the stringent emissions requirements of carbon

border adjustments currently being contemplated around the world, including those which went into force such as the EU Carbon Border Adjustment Mechanism or CBAM (CATF, 2022).

Introduction

Steel is an essential material for global infrastructure and manufacturing, but its production remains one of the most carbon-intensive industrial activities (EIA, 2022). The sector is responsible for approximately 7% of global energy-related CO₂ emissions (IEA, 2023), largely because traditional reduction pathways—such as blast furnace - basic oxygen furnace (BF-BOF) and natural gas-based DRI—rely on fossil-derived carbon to reduce iron ore, leading to significant CO₂ emissions (Energy-transitions, 2025). In the DRI process, natural gas is reformed to produce a mixture of carbon monoxide (CO) and hydrogen (H₂) that chemically reduces iron ore, generating CO₂ as a byproduct (Materials Processing Institute, 2021). This fossil-based chemistry makes the process inherently emissions-intensive, even when high energy efficiency is achieved (Materials Processing Institute, 2021).

Decarbonizing steel production is a central challenge for achieving national and international climate goals (OECD, 2024). Among various technological pathways under consideration—including carbon capture, utilization and storage (CCUS), increased scrap recycling, and direct electrification—hydrogen-based DRI offers the potential to eliminate a significant proportion of Scope 1 process emissions by replacing CO with renewable H₂ as the reductant (Nurdiawati & Urban, 2021). When green hydrogen produced by an electrolyser is used as the reductant, iron ore becomes metallic iron and steam, eliminating CO₂ emissions from the reduction step (Worldsteel, 2025).

This approach has gained momentum globally through projects such as HYBRIT in Sweden and H2 Green Steel, recently renamed Stegra (Sampson, 2023) and is recognized in international collaborations like the Hydrogen Breakthrough Agenda (UNEPFI, 2021) as a priority for near-zero-emissions steel. While the UAE has developed a mature and rapidly expanding renewable energy infrastructure, limited operational data is available globally on hydrogen-based DRI under high ambient temperatures and climatic conditions characteristic of the Middle East and North Africa (MENA).

The MENA region, and particularly the UAE, is uniquely positioned to lead in hydrogen steelmaking due to its high share of global DRI production (Worldsteel, 2022), abundant solar irradiance, and rapidly expanding renewable energy capacity. National strategies further reinforce this opportunity: the UAE National Hydrogen Strategy 2050 (U.ae, 2025a) targets 1.4 million tonnes of hydrogen production annually by 2031, with a phased transition from blue and grey toward green hydrogen powered by solar and wind. Abu Dhabi's Public Policy on Low-Carbon Hydrogen sets out the guiding principles and institutional design for the low-carbon hydrogen sector in Abu Dhabi and it requires all relevant public and commercial entities to develop and implement further coordinated and supporting strategies, policies, and frameworks; and the Abu Dhabi Climate Change Strategy (2023-2027) commits to a 22% emissions reduction by 2027 and sectoral resilience measures (Abu Dhabi Government, 2025). Together these frameworks align with the UAE's Net Zero by 2050 strategic initiative and its commitment to participate in global hydrogen and low-emissions products trade.

This pilot project discussed in this paper aimed to validate the technical feasibility, metallurgical performance, safety, and life-cycle carbon benefits of substituting natural gas with > 99.99 vol% renewably produced hydrogen in an operational DRI. The project also sought to align with emerging regulatory frameworks such as the EU Carbon Border Adjustment Mechanism (CBAM) (European Commission, 2025), ensuring that UAE-produced steel remains competitive in low-carbon export markets.

This paper presents the technical design, operational performance, life-cycle assessment, and lessons learned from the project conducted at a steel manufacturing company. The results demonstrate both the technical readiness of hydrogen-based steelmaking in MENA conditions and its strategic role in advancing the UAE's hydrogen economy, industrial decarbonization, and global trade positioning.

Design Methodology

The MENA region, and especially the Gulf Cooperation Council (GCC) region, has been leading the world in terms of production of steel based on DRI, and rich resources of natural gases coupled with competitive energy infrastructure guarantee a large share of this production ([Worldsteel, 2022](#)). The UAE has focused its investments on energy-intensive businesses, which are currently seeking to decarbonize their process due to the rising climate ambitions. At the same time, the region holds strong prospects for clean energy integration, supported by high solar irradiance levels and rapidly expanding renewable energy capacity. A notable example is Masdar's recently announced pioneering 1 GW round-the-clock (RTC) renewable energy project, which demonstrates the UAE's capability to deliver continuous green power for industrial applications such as hydrogen-based steelmaking.

Few steelmakers around the world are also adopting hydrogen-based technology to achieve their net-zero goals. Other projects, like HYBRIT in Sweden and Stegra, are leading the hydrogen DRI based processes ([Sampson, 2023](#)). Likewise, the Hydrogen Breakthrough Agenda presented at COP26 is considered near-zero emission steel one of the priority zones in international collaboration ([UNEPFI, 2021](#)). The developments given mark the new paradigm of putting green steel in the spotlight of industrial innovation.

The opportunities of adopting hydrogen-based steelmaking early can bring great strategic benefits to MENA economies. It makes the UAE and other countries in the region leaders in global supply chains, with the green steel export to other countries, and establishes a new market in which countries are becoming increasingly mobilized by satisfying the requirements associated with CBAM ([European Commission, 2025](#)). In essence, the pilot project reaffirms the UAE's global sustainability agenda.

The UAE Hydrogen Strategy 2050 was developed as a pillar of the UAE's climate and industrial diversification. The Hydrogen Strategy aims to position the UAE among the first ten global hydrogen producers in the middle of the century, with an aim to produce 1.4 million tonnes annually by 2031 ([U.ae, 2025a](#)). It presents three value chains, including blue, grey, and green hydrogen, with a focus on a gradual transition to green hydrogen with increased usage of solar and wind power ([U.ae, 2025a](#)).

In line with the Net Zero by 2050 strategic initiative, the Hydrogen Strategy roadmap prioritizes domestic industrial applications and use cases for scaling up hydrogen consumption. Emission-intensive steelmaking, in particular, is an excellent opportunity to showcase practical hydrogen applications and use cases. The UAE's participation in international hydrogen initiatives such as the Clean Hydrogen Mission and the Breakthrough Agenda is further evidence that the country adheres to international best practices recognized globally. The pilot project serves as a lighthouse project within this national strategy. The pilot project validated the technical assumptions associated with using green hydrogen in a DRI plant and contributed to supported the UAE's clean energy vision. Its success may also jumpstart the formulation of policy frameworks and mobilizing investments for the development of supporting infrastructure and ecosystems build around aggregating hydrogen demand, which may evolve into clean industrial zones.

The launch of the first green hydrogen pilot project targeting industrial steelmaking in the MENA region was marked in early 2023. The operational platform for the pilot project was an operating DRI plant located in the Industrial City Abu Dhabi (ICAD), serving as the site for the pilot project. The initiative involved producing green hydrogen via electrolyzers powered by solar-derived electricity. The primary objective was to assess the technical feasibility of installing an electrolyzer plant within the existing steel plant to produce green hydrogen and partially replace natural gas in the DRI reactor, a key step in the steelmaking process. This substitution directly targets one of the sector's largest emission sources by eliminating the reliance on carbon-based reducing agents. On-site hydrogen production was achieved using a proton exchange membrane (PEM) electrolyser system integrated with a certified renewable power supply, ensuring that the hydrogen met international standards for green classification through I-REC verification.

The hydrogen produced by the pilot project was certified and audited by third parties. In fact, the pilot project accomplished three industry firsts. First, the certification was in the form of a digital product passport

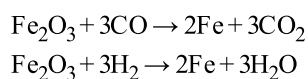
based on the first known global application of the ISO/TC 19870:2023 methodology both location- and market based sub-methodologies (the ISO technical specification on methodologies for determining the greenhouse gas (GHG) emissions associated with the production, conditioning, and transport of hydrogen), which contributed to the standardization of lifecycle assessment methodologies. Second, it was the first to stack and redeem Green Energy Certificates, a form of international renewable energy certificates or I-RECs, with a composite hydrogen certificate, the I-TRACK (HX), to report Scope 2 emissions, demonstrating the interoperability of different product certificates. Third, it was the first to apply a "net-zero benchmark" value on the produced hydrogen, determining where a product sits on the industry-wide "decarbonization scale". The certification data was validated by an internationally recognized assurance service provider.

This paved the way for the pilot project to create early downstream demand for green steel. An agreement between the steel manufacturing company and a first-mover real estate developer was concluded, thus becoming the first such company to commit to the procurement of hydrogen-based steel for construction.

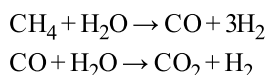
This pilot project featured a modular PEM electrolyser aimed at producing a hydrogen flow up to 42 kg/h and utilise the produced green hydrogen in steel making without retrofitting the direct reduction reactor. The pilot aims to achieve several key objectives. Firstly, it will validate the technical feasibility of integrating megawatt-scale electrolysis with DRI operations under continuous industrial loads. Secondly, the project seeks to achieve greater than 99.9 vol% hydrogen purity through on-site electrolysis and purification, ensuring compatibility with metallurgical specifications. Additionally, the pilot project will focus on evaluating performance by quantifying specific energy consumption (kWh/kg H₂), DRI metallization, reactor efficiency, and in site emissions reductions. Safety and operability were prioritized through the demonstration of leak detection, ventilation, material compatibility, and emergency protection systems under industrial hydrogen service conditions.

The pioneering idea behind the project is to substitute natural gas with green hydrogen in the DRI process. This process is described as traditionally contributing to Scope 1 carbon emissions of steel because of extracting the oxygen in iron ore (usually hematite, Fe₂O₃) to manufacture metallic iron (Fe) ([Worldsteel, 2025](#)). In the traditional methods, this is done by a combination of carbon monoxide (CO) and hydrogen (H₂), obtained through reformed methane (CH₄), and this cannot fail to produce carbon dioxide (CO₂).

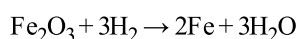
The conventional reaction can be represented as:



Methane reforming to generate reducing gases is itself emissions-intensive:



In the case of reductant hydrogen alone, the only byproduct is water vapor:



In this clean reaction, direct CO₂ emissions from the reduction step are eliminated, resulting in a substantial decrease in Scope 1 emissions ([IEA, 2023](#)). Electricity-related emissions (Scope 2) were fully offset through International Renewable Energy Certificates (I-RECs) sourced from solar power. This approach aligns with the Green Hydrogen Standard, which defines green hydrogen as hydrogen produced using 100 % (or near-100 %) renewable energy and achieving < 1 kg CO₂e per kg H₂ on a 12-month average, thereby meeting internationally recognized criteria for green hydrogen designation ([GH2, 2022](#)).

The on-site production method of the green hydrogen involved proton exchange membrane (PEM) electrolyzers. The reduction reactor into which the project was undertaken was integrated at a very little retrofitting, proving that hydrogen can indeed be used to replace natural gas without any loss in metallurgical production or process stability.

Downstream certification, in accordance with ISO 19870:2023, verified the life-cycle environmental performance and product quality. The certified hydrogen batch achieved a purity of 99.99 vol % at the point of delivery, with an LCA-verified carbon intensity of 0.28 kg CO₂e per kg H₂ (market-based method using I- RECs). During the performance guarantee test, the hydrogen production run delivered 300 kg of H₂ over 8 hours and 19 min at a purity of 99.99 vol %, consuming 16,213 kWh of electricity (specific energy consumption: 54.04 kWh/kg H₂) and 4.95 m³ of desalinated water (specific water consumption: 16.5 L/kg H₂). During the certification test, the pilot operated for 12 h, producing 322 kg of H₂ with an associated electricity consumption of 19,417 kWh. These specifications confirmed compliance with international green hydrogen criteria and established the technical feasibility for large-scale substitution of natural gas in direct reduction ironmaking.

Modular pressurized PEM electrolysis was chosen for the pilot since it was promising but still under maturation and in order to test the Operation and Maintenance challenges of modularized cabinets concept. Key integration insights include dynamic load response, PEM stacks can ramp up and down across a wide operating range from 10% to 100% rapidly, and efficiency optimization. Accounting for the PEM stack and balance of plant (power converters, water treatment, cooling, and drying), the specific energy consumption typically ranges from ~55 to ~70 kWh/kg H₂, depending on the load profile, ambient conditions, degradation, and integration settings. Thermal integration, such as recovering 70-80 °C waste heat from the cooling system to supply low-grade processes (e.g., seawater desalination), could potentially improve the plant efficiency, but its effectiveness depends on plant scale, ambient conditions and site specifics. Purity assurance is achieved via a pressure-swing adsorption (PSA) dryer, delivering >99.99 vol % H₂. Finally, comprehensive maintenance protocols with electrical, pressure, and temperature sensors continuously monitoring system health and stack degradation enable optimal timing for stack replacement. This experience confirms that water electrolysis, when carefully integrated, can provide a robust green-hydrogen supply for industrial DRI while highlighting pathways to enhance flexibility and efficiency through next-generation electrolyser adoption.

Results of the concept

During execution, the project generated actionable insights on green hydrogen production and its integration and utilization in the Direct Reduced Iron (DRI) process. Engineering design feedback was provided to the electrolyser original equipment manufacturer (OEM), informing adaptations of the system for UAE operating conditions (high ambient temperatures, dust, and site utilities). Process safety reviews, including Hazard Identification (HAZID) and Hazard and Operability Study (HAZOP), were completed, with recommendations implemented in the design and procedures. A pre-startup HAZOP verified readiness for commissioning and start-up, confirming interlocks, alarm set-points, leak tests, and emergency procedures. Product assurance was established through calibrated metering and analytics to certify green-hydrogen quality and associated attributes. Operations were continuously monitored and recorded, creating a dataset for performance assessment, troubleshooting, and future optimisation.

Under green hydrogen utilization and production metrics, the electrolyser could deliver up to 42 kg H₂ per hour at an operating pressure of 10 barg. The first green hydrogen was injected in August 2024 to the DRI process. The direct reduction reactor consumed 92 % of the supplied hydrogen, with an 8% purge maintained for safety and O₂ sweep purposes. The hydrogen used in the pilot was certified under ISO 19870:2023, confirming a purity of 99.99 vol % at the point of delivery and an LCA carbon intensity of 0.28 kg CO₂e/kg H₂ (market-based method, 100% renewable electricity via I-RECs sourced from solar power). The hydrogen-based operation achieved an average DRI metallization rate of 97.2 wt % (baseline NG-DRI: 96.8 wt %), with final product quality maintained at 0.2 wt % carbon and silicon and manganese contents within standard metallurgical tolerances. Energy consumption was measured at 54 kWh/kg H₂ (stack + balance of plant) at 0.4 A/cm² for the electrolyser, and 14 GJ/t DRI for the furnace (versus 12 GJ/t for NG-

DRI), reflecting hydrogen's lower volumetric energy density and its impact on heat distribution. Process CO₂ emissions were reduced from 5 t CO₂/t DRI (NG-DRI baseline) to 0.5 t CO₂/t DRI, representing a 90% reduction, while NO_x emissions were measured at 50 ppm (at 3% O₂).

Operation reflected the pilot's learning objectives rather than commercial steady-state. Hydrogen production was dispatched to match DRI demand, and uptime therefore varied, with planned outages for balance-of-plant maintenance and OEM-led stack replacements to install next-generation stacks. The campaign's primary purpose was to test the performance of the system in Abu Dhabi ambient conditions, mature Operations & Maintenance (O&M) practices, validate the modularized cabinet concept, and build operator capability through structured test-and-learn cycles. Safety performance was established through HAZID, HAZOP, dispersion modelling and pre-startup safety reviews that informed detector placement, ventilation and shutdown zoning. During pilot operations there were no reportable hydrogen releases, and design/operational learnings were captured in updated procedures, alarm set-points and training.

During the piloting, iterative engineering design reviews were conducted with the original equipment manufacturer (OEM) to adapt the system to UAE-specific climatic and operating conditions. Process safety was embedded across the project lifecycle: formal HAZID and HAZOP studies were completed, and their recommendations were incorporated into the final design. Immediately prior to commissioning, a pre-startup HAZOP verified operational readiness and confirmed that all required safeguards and procedures for safe start-up were in place.

To validate product integrity, adequate monitoring and measurement systems were installed, enabling certification of the green hydrogen produced. During operation, plant performance was closely monitored and recorded to capture lessons learned and optimize future implementation. Importantly, no unplanned hydrogen releases above 1 vol% were detected, confirming both process safety and system reliability.

Overall, the hydrogen-DRI pathway achieved a 99% reduction in process emissions (approximately 90% when including pellet production and downstream steelmaking). These results, supported by ISO 19870:2023 certification, confirm that green-hydrogen DRI can deliver transformational emission reductions, provided low-carbon electricity is used and effective material recycling strategies are implemented.

Discussion & Conclusion

This study demonstrates that hydrogen-based direct reduction in steel manufacturing, when powered by certified renewable energy, can achieve substantial emissions reductions without compromising metallurgical performance. The pilot project, conducted at a steel manufacturing facility in Abu Dhabi, UAE, successfully integrated a PEM electrolyser pilot with a capacity up to 42 kg H₂ per hour into an operational direct reduction plant, achieving hydrogen purity of 99.99 v% and metallisation rates of 97.2 wt %. Life-cycle assessment results, verified under ISO 19870:2023, indicate a market-based carbon intensity of 0.28 kg CO₂e per kg H₂, translating into a 99% reduction in process emissions compared to natural gas-based DRI.

On the green-hydrogen side, the pilot focused on integrating a new electrolyser under harsh site conditions—high ambient temperatures and dusty atmospheres—while coupling it to an operating DRI plant and aligning controls, operations, and maintenance. On the DRI side, CFD-optimised manifold design was applied to ensure uniform hydrogen flow distribution, and advanced refractory materials were used to mitigate hydrogen embrittlement and thermal cycling effects. Additionally, model-based process control was implemented, whereby furnace operating parameters such as temperature, gas composition, and metallisation rates were dynamically monitored and adjusted using predictive algorithms. This digital control approach enabled stable furnace operation under variable hydrogen load conditions, ensuring continuous production at industrial scale. These changes enabled reliable coupling between the hydrogen supply and the DRI process and generated practical lessons for scale-up.

From a policy and market perspective, the project aligns with international standards and emerging low- emissions trade regulations, demonstrating readiness for compliance with mechanisms such as the EU Carbon Border Adjustment Mechanism. The operational and environmental data generated offers a replicable benchmark for future hydrogen-based steel projects in the MENA region and beyond, where climatic conditions, renewable energy availability, and infrastructure integration present unique challenges.

The research contributes to both academic and industrial discourse by providing empirical data on hydrogen-DRI performance in a hot-climate, high-capacity production setting, a context underrepresented in current literature. These outcomes substantiate the role of green hydrogen as a viable pathway for deep decarbonisation of primary steel production and reinforce the UAE's strategic positioning in the global low- carbon materials market, directly advancing the UAE Net Zero by 2050 (U.ae, 2025b) and National Hydrogen Strategy 2050 objectives (U.ae, 2025a).

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